

Literature Answer to the Kadison–Singer Problem-Family in arXiv:0510024

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that the central Kadison–Singer problem-family surveyed in arXiv:0510024 was answered positively by Marcus–Spielman–Srivastava in arXiv:1306.3969. It is not an original proof from this run.

Original Problem Evidence

Casazza–Fickus–Tremain–Weber [1] present the Kadison–Singer problem and its equivalent formulations in operator theory, frame theory, Hilbert-space theory, Banach-space theory, harmonic analysis, time-frequency analysis, and engineering. The introduction states that the paper reduces these problems to equivalent forms of Kadison–Singer. The source explicitly includes Anderson paving, Weaver’s conjectures, the Feichtinger conjecture, the R_ϵ -conjecture, and the Bourgain–Tzafriri conjecture among the central formulations.

Supporting Answer Evidence

Marcus–Spielman–Srivastava [2] prove two results known to imply a positive solution to Kadison–Singer. First, they prove Weaver’s KS_2 conjecture. Second, they prove Anderson’s paving conjecture in the Casazza–Edidin–Kalra–Paulsen formulation and give explicit paving bounds. Their history and introduction identify the same equivalent conjectures surveyed by [1], including Weaver KS_r , Anderson paving, Bourgain–Tzafriri, Feichtinger, and the R_ϵ -conjecture.

The post-solution survey [3] records the consequence in the language of the 2005 survey: the Marcus–Spielman–Srivastava theorem proves Weaver’s conjecture and therefore yields the Kadison–Singer solution and its standard equivalent consequences.

Literature Answer. *The central Kadison–Singer/Weaver/paving/Feichtinger/Bourgain–Tzafriri open-problem family surveyed in arXiv:0510024 has a positive answer in the later literature.*

Proof. The 2005 source records the equivalence between the Kadison–Singer problem and several finite-dimensional partition formulations, in particular Anderson paving and Weaver-type vector partition conjectures. Marcus–Spielman–Srivastava prove Weaver KS_2 , which Weaver had shown sufficient for Kadison–Singer, and they also prove Anderson paving directly with explicit bounds. These two theorems give the positive Kadison–Singer solution and hence settle the equivalent conjectures in the scope of the 2005 survey. $\square$

Limitations

This packet only records the later literature answer to the central problem-family of arXiv:0510024. It does not claim new proof, improved paving constants, or a classification of every special operator class and uniform paving variant discussed in the survey. It also does not address unrelated Banach-space generalizations of Feichtinger-type conjectures.

Verification Notes

The lane-15 queue selected arXiv:0510024. The cheap run indexes had no exact previous packet for this source. Searches for Kadison–Singer, Paving, Weaver, Feichtinger, Bourgain–Tzafriri, and Anderson identified nearby later Kadison–Singer sources, especially arXiv:1306.3969 and arXiv:1407.4768. A web check on 2026-07-03 confirmed the arXiv record for arXiv:1306.3969.

References

- [1] P. G. Casazza, M. Fickus, J. C. Tremain, and E. Weber, *The Kadison–Singer Problem in Mathematics and Engineering: A Detailed Account*, arXiv:math/0510024.
- [2] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing Families II: Mixed Characteristic Polynomials and the Kadison–Singer Problem*, arXiv:1306.3969.
- [3] P. G. Casazza, *Consequences of the Marcus/Spielman/Srivastava Solution of the Kadison–Singer Problem*, arXiv:1407.4768.